

Kapsamli Model Karsilastirma Raporu v2

Genetik Varyant Patojenite Tahmini

6 Model Tipi | 5 FE Modu | 2 Egitim Modu
96 Toplam Konfigurasyon | Optuna Bayesian Optimizasyon

Modeller: LightGBM, XGBoost, Neural Net, Deep NN, Stacking Ensemble
Veri Seti: Genetik Varyant | 4 Gen Paneli

YENILIKLER: Full-FE (In-silico + Gelismis FE), Focal Loss,
Delta K-mer, Yonlu BLOSUM62, Conservation PCA, Stacking Ensemble

En Iyi Sonuc: XGBoost
F1 = 0.8148 | AUC-ROC = 0.9253
FE Modu: With FE (leaked)

Olusturulma: 2026-03-22 17:02

1. Metodoloji

1.1 Deneyisel Tasarım

4 model mimarisi (LightGBM, XGBoost, Neural Net, Deep NN), 4 feature engineering modu (no_fe, with_fe, clean, clean_fe) ve 2 eğitim modu (single-panel, multi-panel) kullanılarak kapsamlı bir deneyisel karşılaştırma yapılmıştır. Ek olarak 2 stacking ensemble konfigürasyonu (LR ve LightGBM meta-learner) test edilmiştir.

v1 (leaked) sonuçları: Orijinal eğitim ~60 meta-predictor skorunu (REVEL, CADD, ClinPred, AlphaMissense, MetaSVM vb.) feature olarak kullanıyordu. Bu skorlar ClinVar üzerinde eğitilmiş ML modelleri tarafından üretilmiştir -- aynı ClinVar verisi üzerinde bu skorları kullanmak döngüsellik (circularity) yaratır ve gerçekçi olmayan yüksek performans üretir.

v2 (clean) sonuçları: Tüm meta-predictor skorları, rankscore değerleri, prediction kategorileri, gnomAD populasyon frekansları ve tool_consensus tamamen kaldırıldı. Sadece biyolojik olarak anlamlı özellikler korundu: evrimsel korunma (GERP, PhyloP, PhastCons), fizikokimyasal delta değerleri ve sekans bilgisi.

1.2 Modeller

LightGBM: Yaprak-öncelikli ağac büyümesi, 200 Optuna trial, 5-fold CV. Clean modlarda Focal Loss opsiyonel (gamma ve alpha Optuna ile tune edilir).

XGBoost: Seviye-öncelikli ağac büyümesi, L1/L2 regularizasyon, 200 Optuna trial, 5-fold CV. Clean modlarda Focal Loss desteği.

Neural Network (NN): 1-2 katmanlı MLP, BatchNorm + ReLU + Dropout. PyTorch + DirectML (AMD GPU). 100 Optuna trial, 3-fold CV. Clean modlarda FocalLoss kullanılır.

Deep Neural Network (DNN): 3-5 katmanlı MLP, her 2 katmanda residual bağlantı. Dropout (0.2-0.6), ReduceLROnPlateau. 100 Optuna trial.

Stacking Ensemble: Base modeller (LightGBM + XGBoost + SklearnMLP) ile StackingClassifier. Meta-learner olarak LogisticRegression (stacking_lr) veya LightGBM (stacking_lgbm). StratifiedKFold(5) CV ile stacking leakage önlenir.

1.3 Feature Engineering Modları

no_fe (leaked baseline): Orijinal 99 numerik/kategorik feature + 656 k-mer (CountVectorizer, max_features=200). Meta-predictor skorları dahil -- veri sızıntısı var.

with_fe (leaked FE): Grantham, BLOSUM62, CpG, GC content, gnomAD türleri, tool_consensus. ~57 feature. Meta-predictor bazlı özellikler -- veri sızıntısı var.

clean (temiz baseline): Tüm meta-predictor, rankscore, gnomAD ve prediction sütunları kaldırıldı. İyileştirilmiş k-mer (max_features=500, min_df=2). Sızıntı yok.

clean_fe (temiz + gelişmiş FE): Clean modun ustune eklenen gelişmiş özellikler: Yönlü BLOSUM62 (ref_self, alt_self, delta), Conservation PCA (4 skor -> 2 PC), GERP x PhyloP etkileşimi, Delta K-mer (Ref-Alt farkı), Grantham mesafesi/kategorisi, CpG/GC/transition. Korelasyon filtresi ($r > 0.99$). Sızıntı yok.

full_fe (in-silico + gelişmiş FE): In-silico meta-predictor skorları KORUNARAK gelişmiş FE özellikleri eklenmiştir. Yarışma veri setinde in-silico skorlar sağlanacağı için bu mod en kapsamlı feature setini sunar. Tüm clean_fe özellikleri + REVEL, CADD, ClinPred, AlphaMissense, MetaSVM, SIFT vb. skorlar + gnomAD türleri dahildir.

1.4 Focal Loss

Sınıf dengesizliğini (68% benign, 32% pathogenic) ele almak için Focal Loss kullanılmıştır. $FL = -\alpha * (1-p)^\gamma * \log(p)$. gamma parametresi kolay örneklerin ağırlığını azaltır, alpha sınıf dengesini ayarlar. Clean modlarda otomatik uygulanır, gamma ve alpha Optuna ile optimize edilir.

2. Veri Sızıntısı (Data Leakage) Analizi

Orijinal modeller (v1), ClinVar üzerinde eğitilmiş ~60 meta-predictor skorunu feature olarak kullanıyordu. Bu skorlar (REVEL, CADD, ClinPred, AlphaMissense, MetaSVM, MetaLR, MetaRNN, ESM1b, MiSTIC, MutationTaster, SIFT, PROVEAN, VEST, BayesDel, FATHMM, CHASMplus, vb.) kendi basına ClinVar verisi üzerinde eğitilmiş ML modelleridir.

Bu dongsellik (circularity) nedeniyle v1 modelleri gerçekçi olmayan yüksek performans gösteriyordu. v2'de bu skorların tamamı kaldırıldı ve sadece biyolojik olarak anlamlı özellikler (evrimsel korunma, fizikokimyasal özellikler, sekans bilgisi) kullanıldı.

Performans Karsılastırması

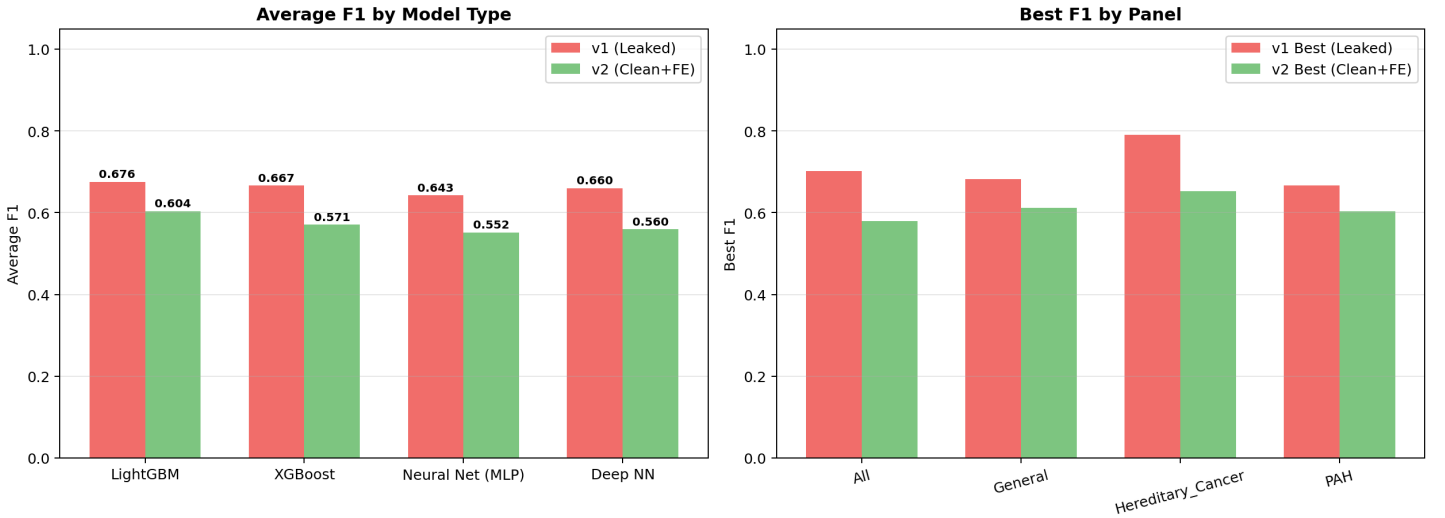
v1 (leaked) ortalama F1: 0.6615
v2 (clean_fe) ortalama F1: 0.5716
F1 düşüşü: 0.0899

v1'den v2'ye geçiste ortalama F1 skoru 0.0899 düşmüştür. Bu beklenen bir sonuctur -- yüksek v1 performansı meta-predictor circularity'den kaynaklanıyordu. v2 sonuçları gerçek model performansını yansıtmaktadır.

Model Bazında v1 vs v2 Karsılastırma

LightGBM: v1=0.6758 v2=0.6039 (delta=-0.0719)
XGBoost: v1=0.6666 v2=0.5705 (delta=-0.0960)
Neural Net (MLP): v1=0.6431 v2=0.5521 (delta=-0.0910)
Deep NN: v1=0.6604 v2=0.5598 (delta=-0.1006)

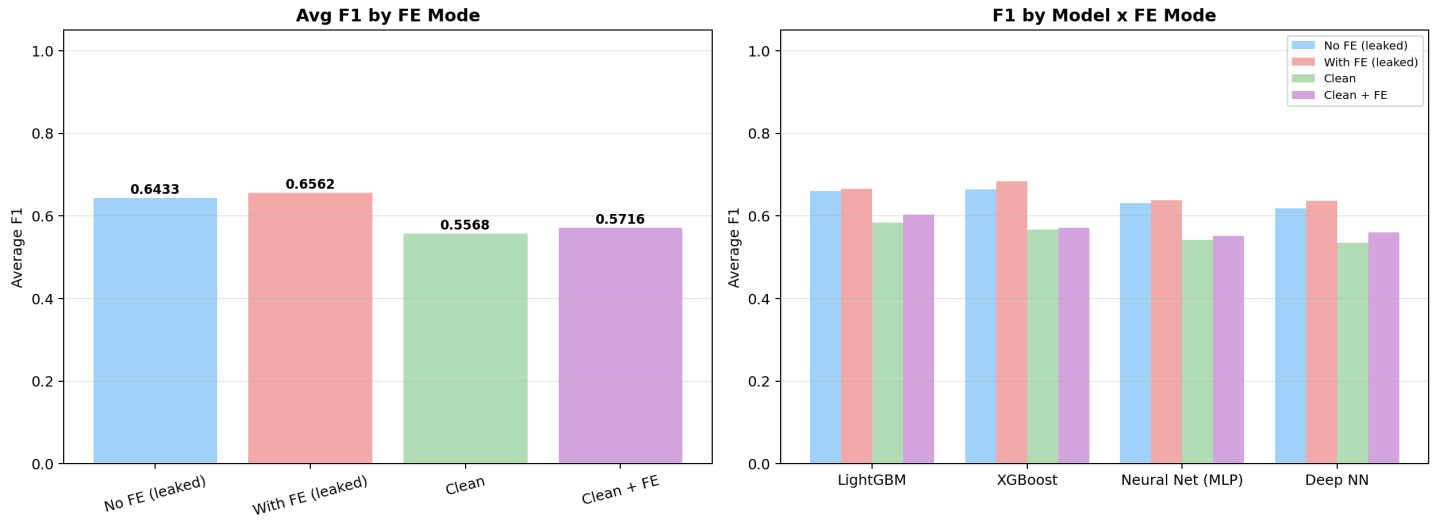
Data Leakage Impact: v1 (Leaked) vs v2 (Clean)



3. Feature Engineering Modu Karsilastirmasi

No FE (leaked): Avg F1=0.6433 Best F1=0.7949 ~531 feat Sizinti: EVET
 With FE (leaked): Avg F1=0.6562 Best F1=0.8148 ~216 feat Sizinti: EVET
 Clean: Avg F1=0.5568 Best F1=0.6222 ~1053 feat Sizinti: HAYIR
 Clean + FE: Avg F1=0.5715 Best F1=0.6526 ~1496 feat Sizinti: HAYIR
 Full FE (in-silico + FE): Avg F1=0.6425 Best F1=0.7952 ~1530 feat Sizinti: HAYIR

Feature Engineering Mode Comparison (v2)

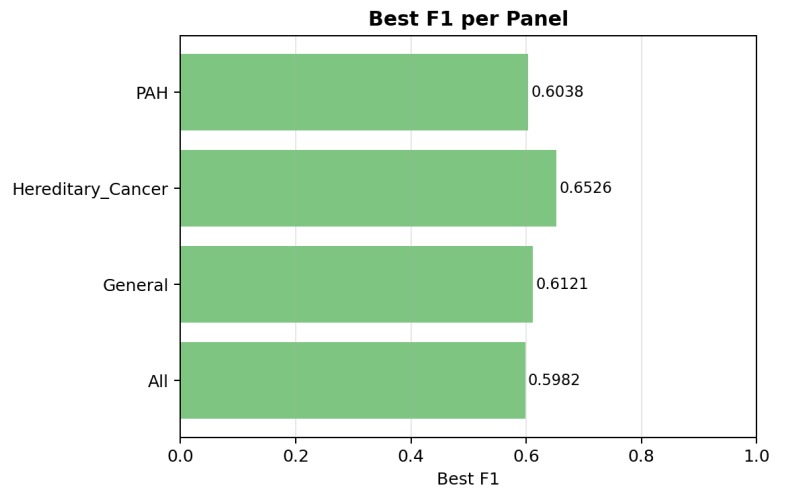
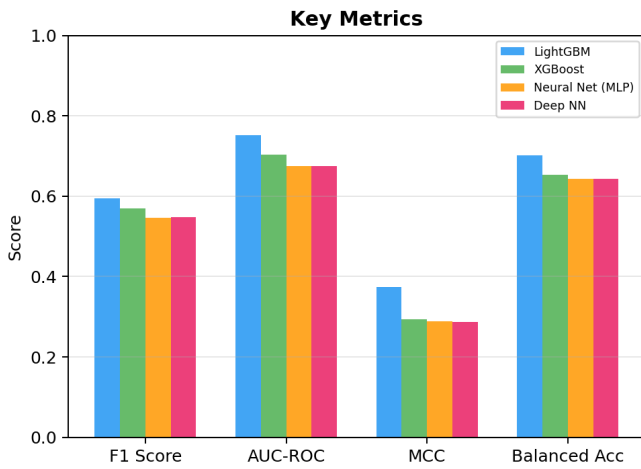
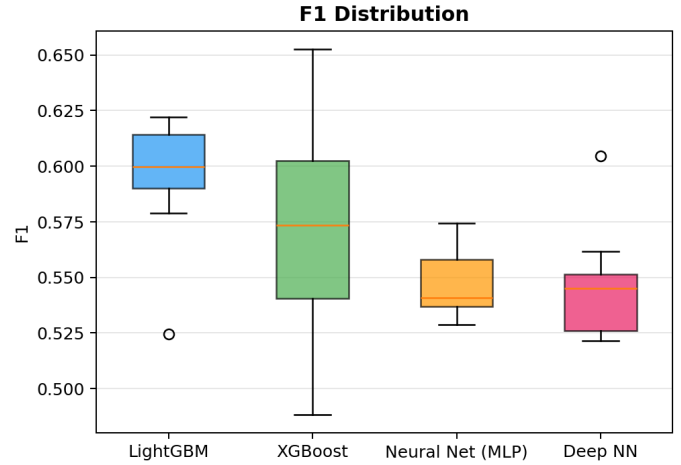
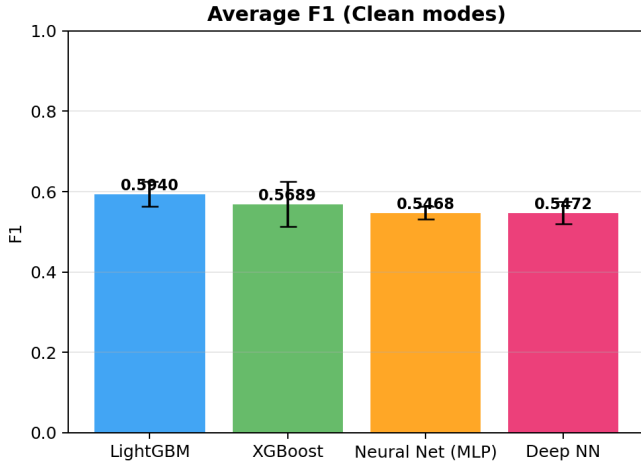


4. Temiz Model Analizi (Sizintisiz)

Asagidaki analizler sadece clean ve clean_fe FE modlarini kapsamaktadır. Bu modlarda hicbir meta-predictor skoru, rankscore veya gnomAD degeri kullanilmamistir.

LightGBM: Avg F1=0.5940 Best F1=0.6222 (n=8)
 XGBoost: Avg F1=0.5689 Best F1=0.6526 (n=8)
 Neural Net (MLP): Avg F1=0.5468 Best F1=0.5743 (n=8)
 Deep NN: Avg F1=0.5472 Best F1=0.6047 (n=8)

Clean Model Analysis (No Data Leakage)



5. Stacking Ensemble Sonuclari

Stacking ensemble, 3 base model (LightGBM, XGBoost, SklearnMLP) ciktilarini birlestirerek daha guclu bir tahmin uretir. StratifiedKFold(5) CV ile base model tahminleri uretilir, meta-learner bu tahminler uzerinde egitilir.

Stacking (LR): Avg F1=0.5770 Best F1=0.6400

Stacking (LGBM): Avg F1=0.5657 Best F1=0.6306

Tek Model vs Stacking

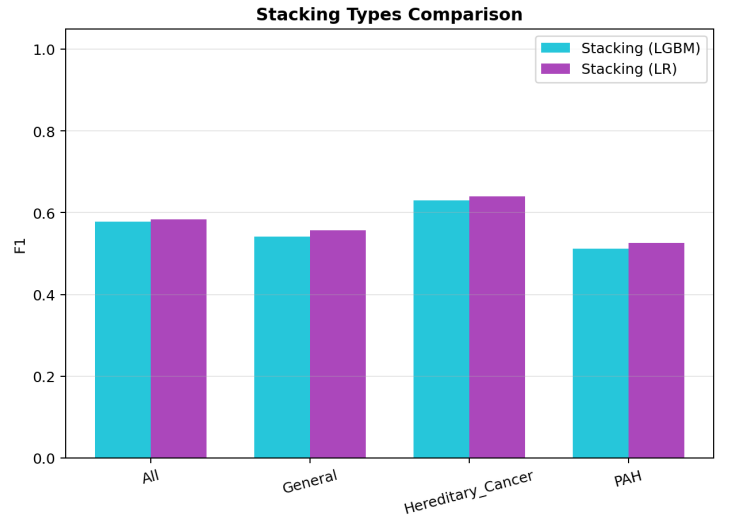
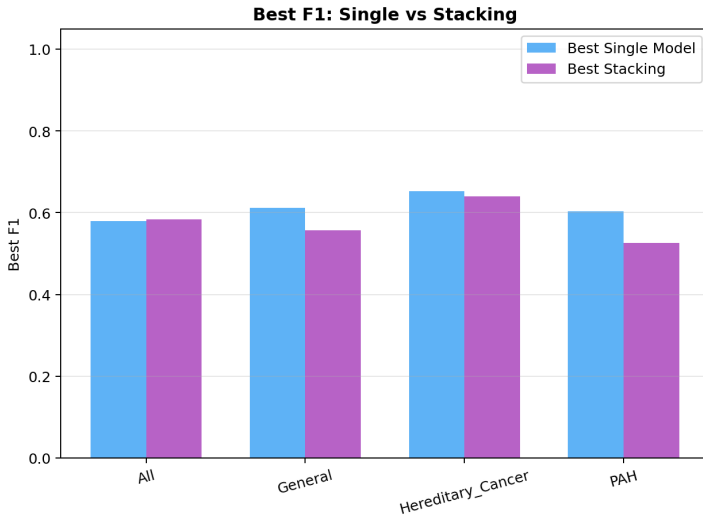
En iyi tek model (clean_fe): F1=0.6526

En iyi stacking: F1=0.6400

Stacking kazanci: -0.0126

#	Model	Panel	Mode	F1	AUC-ROC	MCC	Prec	Recall	Spec	Thresh
1	Stacking (LR)	Hereditary_Cancer	Single	0.6400	0.8114	0.4878	0.5246	0.8205	0.7212	0.36
2	Stacking (LGBM)	Hereditary_Cancer	Single	0.6306	0.8176	0.4825	0.4861	0.8974	0.6442	0.49
3	Stacking (LR)	All	Multi	0.5842	0.7489	0.3392	0.4374	0.8796	0.4692	0.38
4	Stacking (LGBM)	All	Multi	0.5789	0.7316	0.3265	0.4388	0.8504	0.4897	0.46
5	Stacking (LR)	General	Single	0.5575	0.7248	0.3011	0.4222	0.8205	0.4989	0.40
6	Stacking (LGBM)	General	Single	0.5412	0.6930	0.2697	0.3884	0.8923	0.3730	0.42
7	Stacking (LR)	PAH	Single	0.5263	0.6028	0.2230	0.4167	0.7143	0.5227	0.46
8	Stacking (LGBM)	PAH	Single	0.5122	0.5087	0.1769	0.3443	1.0000	0.0909	0.13

Single Model vs Stacking Ensemble



6. Full-FE Analizi (In-silico + Gelismis FE)

Full-FE modu, yarisma veri setinde in-silico skorlari saglanacagi goz onunde bulundurularak tasarlanmistir. Meta-predictor skorlari (REVEL, CADD, ClinPred, AlphaMissense, MetaSVM vb.) korunurken, gelismis FE ozellikleri (yonlu BLOSUM62, Conservation PCA, delta k-mer, Grantham, gnomAD turevleri) eklenmistir.

Model Bazinda Full-FE Sonuclari

LightGBM: Avg F1=0.6703 Best F1=0.7595
 XGBoost: Avg F1=0.6620 Best F1=0.7750
 Neural Net (MLP): Avg F1=0.6149 Best F1=0.6667
 Deep NN: Avg F1=0.5962 Best F1=0.6977
 Stacking (LR): Avg F1=0.6684 Best F1=0.7952
 Stacking (LGBM): Avg F1=0.6433 Best F1=0.7048

FE Modu Karsilastirmasi (Ortalama F1)

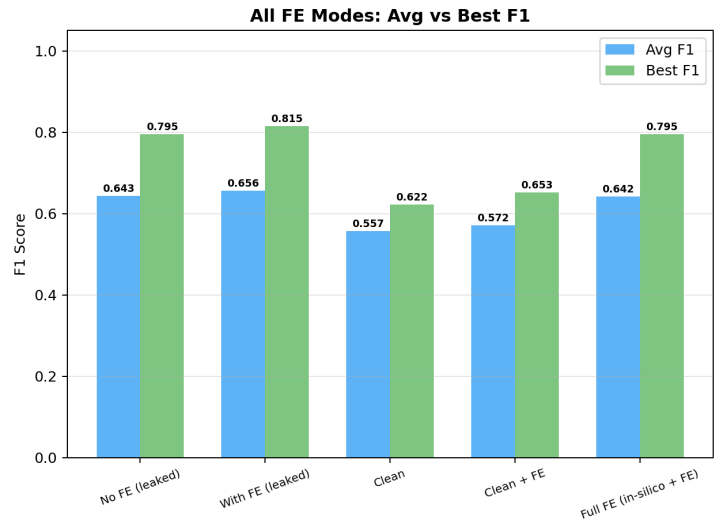
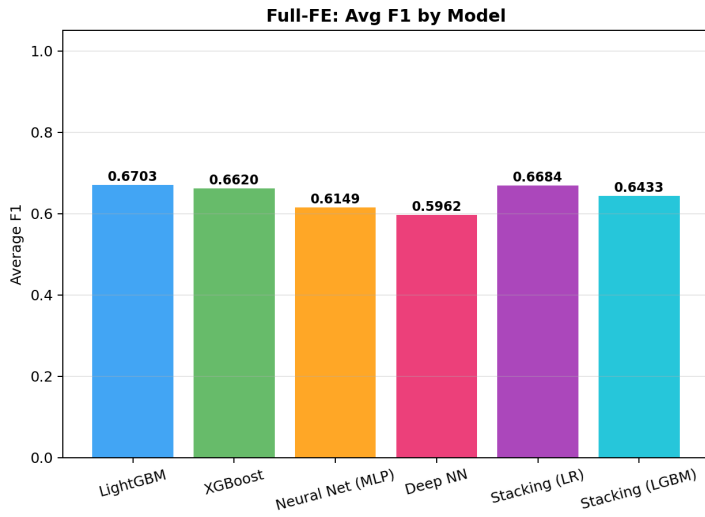
No FE (leaked): Avg=0.6433 Best=0.7949
 With FE (leaked): Avg=0.6562 Best=0.8148
 Clean: Avg=0.5568 Best=0.6222
 Clean + FE: Avg=0.5715 Best=0.6526
 Full FE (in-silico + FE): Avg=0.6425 Best=0.7952

Panel Bazinda Full-FE En Iyi Sonuclar

All: F1=0.6967 Model=LightGBM
 General: F1=0.6637 Model=LightGBM
 Hereditary_Cancer: F1=0.7952 Model=Stacking (LR)
 PAH: F1=0.5882 Model=Neural Net (MLP)

#	Model	Panel	Mode	F1	AUC-ROC	MCC	Prec	Recall	Spec	Thresh
1	Stacking (LR)	Hereditary_Cancer	Single	0.7952	0.9255	0.7144	0.7500	0.8462	0.8942	0.59
2	XGBoost	Hereditary_Cancer	Single	0.7750	0.9126	0.6881	0.7561	0.7949	0.9038	0.54
3	LightGBM	Hereditary_Cancer	Single	0.7595	0.9186	0.6678	0.7500	0.7692	0.9038	0.52
4	Stacking (LGBM)	Hereditary_Cancer	Single	0.7048	0.8624	0.5984	0.5606	0.9487	0.7212	0.25
5	Deep NN	Hereditary_Cancer	Single	0.6977	0.8642	0.5743	0.6383	0.7692	0.8365	0.51
6	LightGBM	All	Multi	0.6967	0.8721	0.5402	0.6411	0.7628	0.7997	0.44
7	Stacking (LR)	All	Multi	0.6918	0.8567	0.5284	0.6077	0.8029	0.7568	0.53
8	XGBoost	All	Multi	0.6879	0.8546	0.5213	0.5881	0.8285	0.7277	0.54
9	Stacking (LGBM)	All	Multi	0.6793	0.8511	0.5086	0.5989	0.7847	0.7534	0.56
10	Neural Net (MLP)	Hereditary_Cancer	Single	0.6667	0.8740	0.5417	0.6667	0.6667	0.8750	0.60
11	LightGBM	General	Single	0.6637	0.8381	0.4935	0.5866	0.7641	0.7597	0.30
12	Stacking (LR)	General	Single	0.6411	0.8151	0.4536	0.5282	0.8154	0.6751	0.42
13	XGBoost	General	Single	0.6395	0.8282	0.4526	0.5498	0.7641	0.7208	0.51
14	Stacking (LGBM)	General	Single	0.6358	0.8132	0.4445	0.5232	0.8103	0.6705	0.55
15	Neural Net (MLP)	All	Multi	0.6121	0.7656	0.4075	0.4502	0.9562	0.4521	0.50
16	Deep NN	All	Multi	0.6000	0.7393	0.3723	0.4505	0.8978	0.4863	0.49
17	Neural Net (MLP)	General	Single	0.5924	0.7409	0.3865	0.4296	0.9538	0.4348	0.50
18	Neural Net (MLP)	PAH	Single	0.5882	0.6645	0.3540	0.4255	0.9524	0.3864	0.51
19	Deep NN	General	Single	0.5870	0.7140	0.3811	0.4209	0.9692	0.4050	0.50
20	LightGBM	PAH	Single	0.5614	0.6526	0.2892	0.4444	0.7619	0.5455	0.39
21	Stacking (LGBM)	PAH	Single	0.5532	0.6591	0.3089	0.5000	0.6190	0.7045	0.42
22	XGBoost	PAH	Single	0.5455	0.6494	0.3144	0.5217	0.5714	0.7500	0.43
23	Stacking (LR)	PAH	Single	0.5455	0.6775	0.2645	0.4412	0.7143	0.5682	0.35
24	Deep NN	PAH	Single	0.5000	0.4708	0.1231	0.3333	1.0000	0.0455	0.50

Full-FE Analysis: In-silico Scores + Advanced FE



7. Tum Sonuclar - Siralı Tablo

Toplam 96 konfigürasyon, F1 skoruna göre sıralanmıştır. Üst 3 yeşil, alt 3 kırmızı ile işaretlenmiştir.

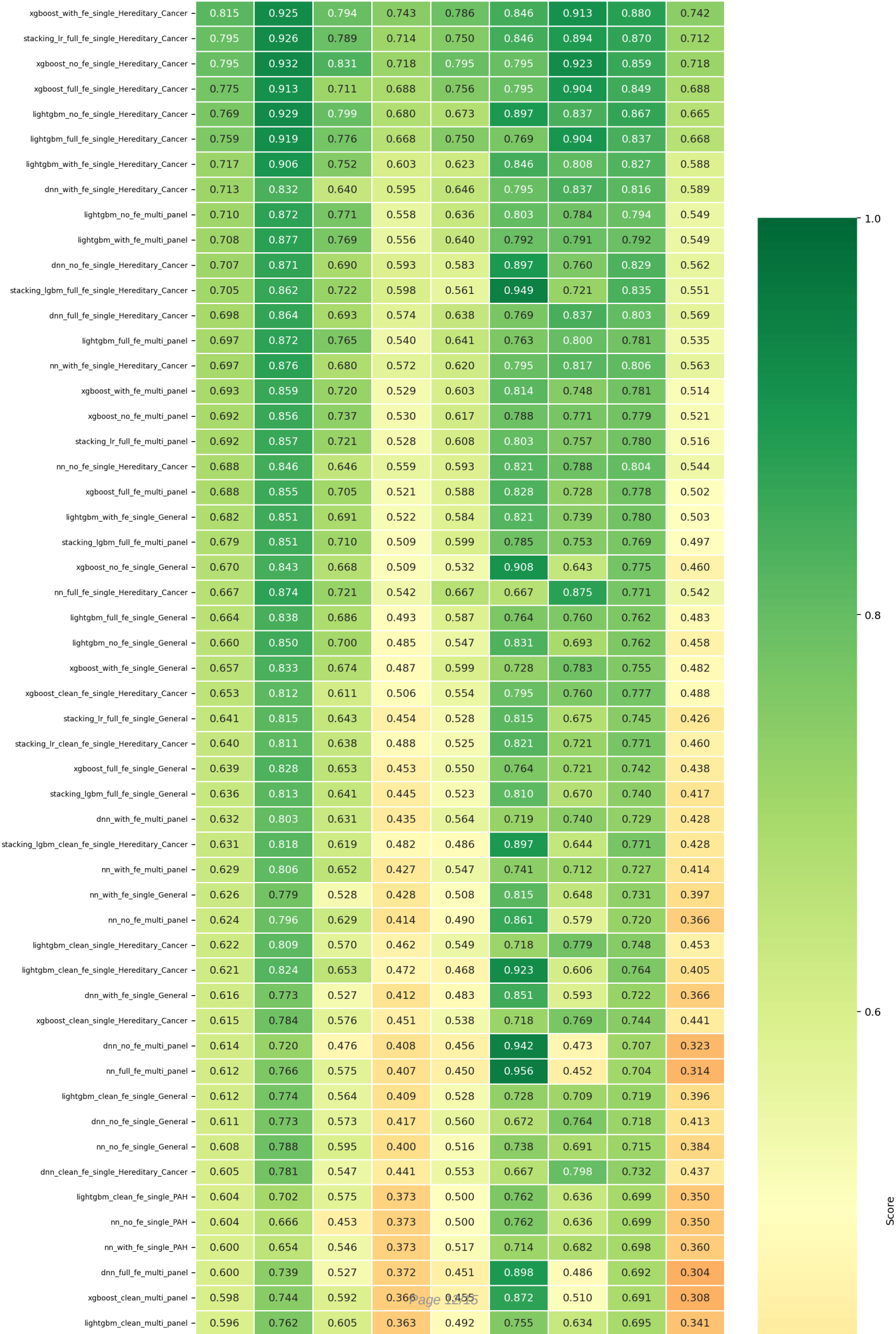
#	Model	Panel	Mode	FE	F1	AUC-ROC	MCC	Prec	Recall	Spec	Thresh
1	XGBoost	Hereditary_Cancer	Single	With FE (leak)	0.8148	0.9253	0.7428	0.7857	0.8462	0.9135	0.43
2	Stacking (LR)	Hereditary_Cancer	Single	Full FE (in)	0.7952	0.9255	0.7144	0.7500	0.8462	0.8942	0.59
3	XGBoost	Hereditary_Cancer	Single	No FE (leak)	0.7949	0.9320	0.7179	0.7949	0.7949	0.9231	0.55
4	XGBoost	Hereditary_Cancer	Single	Full FE (in)	0.7750	0.9126	0.6881	0.7561	0.7949	0.9038	0.54
5	LightGBM	Hereditary_Cancer	Single	No FE (leak)	0.7692	0.9285	0.6795	0.6731	0.8974	0.8365	0.47
6	LightGBM	Hereditary_Cancer	Single	Full FE (in)	0.7595	0.9186	0.6678	0.7500	0.7692	0.9038	0.52
7	LightGBM	Hereditary_Cancer	Single	With FE (leak)	0.7174	0.9058	0.6029	0.6226	0.8462	0.8077	0.24
8	Deep NN	Hereditary_Cancer	Single	With FE (leak)	0.7126	0.8323	0.5955	0.6458	0.7949	0.8365	0.64
9	LightGBM	All	Multi	No FE (leak)	0.7097	0.8719	0.5580	0.6358	0.8029	0.7842	0.53
10	LightGBM	All	Multi	With FE (leak)	0.7080	0.8769	0.5561	0.6401	0.7920	0.7911	0.48
11	Deep NN	Hereditary_Cancer	Single	No FE (leak)	0.7071	0.8711	0.5930	0.5833	0.8974	0.7596	0.63
12	Stacking (LGBM)	Hereditary_Cancer	Single	Full FE (in)	0.7048	0.8624	0.5984	0.5606	0.9487	0.7212	0.25
13	Deep NN	Hereditary_Cancer	Single	Full FE (in)	0.6977	0.8642	0.5743	0.6383	0.7692	0.8365	0.51
14	LightGBM	All	Multi	Full FE (in)	0.6967	0.8721	0.5402	0.6411	0.7628	0.7997	0.44
15	Neural Net (MLP)	Hereditary_Cancer	Single	With FE (leak)	0.6966	0.8762	0.5717	0.6200	0.7949	0.8173	0.61
16	XGBoost	All	Multi	With FE (leak)	0.6925	0.8591	0.5292	0.6027	0.8139	0.7483	0.42
17	XGBoost	All	Multi	No FE (leak)	0.6923	0.8562	0.5302	0.6171	0.7883	0.7705	0.49
18	Stacking (LR)	All	Multi	Full FE (in)	0.6918	0.8567	0.5284	0.6077	0.8029	0.7568	0.53
19	Neural Net (MLP)	Hereditary_Cancer	Single	No FE (leak)	0.6882	0.8464	0.5594	0.5926	0.8205	0.7885	0.50
20	XGBoost	All	Multi	Full FE (in)	0.6879	0.8546	0.5213	0.5881	0.8285	0.7277	0.54
21	LightGBM	General	Single	With FE (leak)	0.6823	0.8505	0.5216	0.5839	0.8205	0.7391	0.39
22	Stacking (LGBM)	All	Multi	Full FE (in)	0.6793	0.8511	0.5086	0.5989	0.7847	0.7534	0.56
23	XGBoost	General	Single	No FE (leak)	0.6705	0.8425	0.5095	0.5315	0.9077	0.6430	0.30
24	Neural Net (MLP)	Hereditary_Cancer	Single	Full FE (in)	0.6667	0.8740	0.5417	0.6667	0.6667	0.8750	0.60
25	LightGBM	General	Single	Full FE (in)	0.6637	0.8381	0.4935	0.5866	0.7641	0.7597	0.30
26	LightGBM	General	Single	No FE (leak)	0.6599	0.8496	0.4852	0.5473	0.8308	0.6934	0.40
27	XGBoost	General	Single	With FE (leak)	0.6574	0.8332	0.4874	0.5992	0.7282	0.7826	0.57
28	XGBoost	Hereditary_Cancer	Single	Clean + FE	0.6526	0.8124	0.5059	0.5536	0.7949	0.7596	0.43
29	Stacking (LR)	General	Single	Full FE (in)	0.6411	0.8151	0.4536	0.5282	0.8154	0.6751	0.42
30	Stacking (LR)	Hereditary_Cancer	Single	Clean + FE	0.6400	0.8114	0.4878	0.5246	0.8205	0.7212	0.36
31	XGBoost	General	Single	Full FE (in)	0.6395	0.8282	0.4526	0.5498	0.7641	0.7208	0.51
32	Stacking (LGBM)	General	Single	Full FE (in)	0.6358	0.8132	0.4445	0.5232	0.8103	0.6705	0.55
33	Deep NN	All	Multi	With FE (leak)	0.6324	0.8030	0.4354	0.5645	0.7190	0.7397	0.55
34	Stacking (LGBM)	Hereditary_Cancer	Single	Clean + FE	0.6306	0.8176	0.4825	0.4861	0.8974	0.6442	0.49
35	Neural Net (MLP)	All	Multi	With FE (leak)	0.6295	0.8063	0.4265	0.5472	0.7409	0.7123	0.50
36	Neural Net (MLP)	General	Single	With FE (leak)	0.6260	0.7791	0.4277	0.5080	0.8154	0.6476	0.45
37	Neural Net (MLP)	All	Multi	No FE (leak)	0.6243	0.7960	0.4135	0.4896	0.8613	0.5788	0.41
38	LightGBM	Hereditary_Cancer	Single	Clean	0.6222	0.8092	0.4619	0.5490	0.7179	0.7788	0.45
39	LightGBM	Hereditary_Cancer	Single	Clean + FE	0.6207	0.8237	0.4725	0.4675	0.9231	0.6058	0.20
40	Deep NN	General	Single	With FE (leak)	0.6160	0.7726	0.4117	0.4826	0.8513	0.5927	0.48
41	XGBoost	Hereditary_Cancer	Single	Clean	0.6154	0.7843	0.4510	0.5385	0.7179	0.7692	0.53
42	Deep NN	All	Multi	No FE (leak)	0.6143	0.7203	0.4076	0.4558	0.9416	0.4726	0.53
43	Neural Net (MLP)	All	Multi	Full FE (in)	0.6121	0.7656	0.4075	0.4502	0.9562	0.4521	0.50
44	LightGBM	General	Single	Clean + FE	0.6121	0.7741	0.4088	0.5279	0.7282	0.7094	0.43
45	Deep NN	General	Single	No FE (leak)	0.6107	0.7730	0.4171	0.5598	0.6718	0.7643	0.50
46	Neural Net (MLP)	General	Single	No FE (leak)	0.6076	0.7875	0.3995	0.5161	0.7385	0.6911	0.48
47	Deep NN	Hereditary_Cancer	Single	Clean + FE	0.6047	0.7811	0.4406	0.5532	0.6667	0.7981	0.85

48	LightGBM	PAH	Single	Clean + FE	0.6038	0.7024	0.3725	0.5000	0.7619	0.6364	0.29
49	Neural Net (MLP)	PAH	Single	No FE (leak)	0.6038	0.6656	0.3725	0.5000	0.7619	0.6364	0.50
50	Neural Net (MLP)	PAH	Single	With FE (leak)	0.6000	0.6537	0.3726	0.5172	0.7143	0.6818	0.49
51	Deep NN	All	Multi	Full FE (in)	0.6000	0.7393	0.3723	0.4505	0.8978	0.4863	0.49
52	XGBoost	All	Multi	Clean	0.5982	0.7445	0.3660	0.4552	0.8723	0.5103	0.56
53	LightGBM	All	Multi	Clean	0.5957	0.7620	0.3628	0.4917	0.7555	0.6336	0.34
54	LightGBM	General	Single	Clean	0.5939	0.7695	0.3709	0.4740	0.7949	0.6064	0.24
55	Neural Net (MLP)	General	Single	Full FE (in)	0.5924	0.7409	0.3865	0.4296	0.9538	0.4348	0.50
56	Deep NN	PAH	Single	With FE (leak)	0.5882	0.7013	0.3503	0.5000	0.7143	0.6591	0.51
57	Neural Net (MLP)	PAH	Single	Full FE (in)	0.5882	0.6645	0.3540	0.4255	0.9524	0.3864	0.51
58	Deep NN	General	Single	Full FE (in)	0.5870	0.7140	0.3811	0.4209	0.9692	0.4050	0.50
59	Stacking (LR)	All	Multi	Clean + FE	0.5842	0.7489	0.3392	0.4374	0.8796	0.4692	0.38
60	XGBoost	All	Multi	Clean + FE	0.5796	0.7423	0.3279	0.4396	0.8504	0.4914	0.49
61	LightGBM	All	Multi	Clean + FE	0.5790	0.7573	0.3274	0.4551	0.7956	0.5531	0.30
62	Stacking (LGBM)	All	Multi	Clean + FE	0.5789	0.7316	0.3265	0.4388	0.8504	0.4897	0.46
63	Neural Net (MLP)	Hereditary_Cancer	Single	Clean + FE	0.5743	0.7147	0.3831	0.4677	0.7436	0.6827	0.58
64	XGBoost	PAH	Single	With FE (leak)	0.5714	0.6721	0.3669	0.5714	0.5714	0.7955	0.37
65	XGBoost	General	Single	Clean	0.5673	0.7275	0.3258	0.4147	0.8974	0.4348	0.54
66	Neural Net (MLP)	All	Multi	Clean + FE	0.5627	0.7119	0.2975	0.4075	0.9088	0.3801	0.50
67	Deep NN	All	Multi	Clean + FE	0.5617	0.6824	0.2913	0.4288	0.8139	0.4914	0.53
68	LightGBM	PAH	Single	Full FE (in)	0.5614	0.6526	0.2892	0.4444	0.7619	0.5455	0.39
69	Stacking (LR)	General	Single	Clean + FE	0.5575	0.7248	0.3011	0.4222	0.8205	0.4989	0.40
70	Neural Net (MLP)	All	Multi	Clean	0.5566	0.6996	0.2950	0.3920	0.9599	0.3014	0.50
71	XGBoost	General	Single	Clean + FE	0.5561	0.7055	0.3005	0.4071	0.8769	0.4302	0.35
72	LightGBM	PAH	Single	With FE (leak)	0.5532	0.6234	0.3089	0.5000	0.6190	0.7045	0.27
73	Stacking (LGBM)	PAH	Single	Full FE (in)	0.5532	0.6591	0.3089	0.5000	0.6190	0.7045	0.42
74	Deep NN	PAH	Single	Clean + FE	0.5479	0.6483	0.2632	0.3846	0.9524	0.2727	0.50
75	Deep NN	All	Multi	Clean	0.5471	0.6955	0.2598	0.3948	0.8905	0.3596	0.50
76	Stacking (LR)	PAH	Single	Full FE (in)	0.5455	0.6775	0.2645	0.4412	0.7143	0.5682	0.35
77	XGBoost	PAH	Single	Full FE (in)	0.5455	0.6494	0.3144	0.5217	0.5714	0.7500	0.43
78	Deep NN	Hereditary_Cancer	Single	Clean	0.5432	0.6908	0.3636	0.5238	0.5641	0.8077	0.51
79	Neural Net (MLP)	Hereditary_Cancer	Single	Clean	0.5421	0.7202	0.3287	0.4265	0.7436	0.6250	0.48
80	Stacking (LGBM)	General	Single	Clean + FE	0.5412	0.6930	0.2697	0.3884	0.8923	0.3730	0.42
81	Deep NN	PAH	Single	No FE (leak)	0.5405	0.6320	0.2439	0.3774	0.9524	0.2500	0.51
82	Neural Net (MLP)	General	Single	Clean + FE	0.5398	0.6596	0.2642	0.3952	0.8513	0.4188	0.50
83	Neural Net (MLP)	PAH	Single	Clean	0.5385	0.6494	0.2624	0.4516	0.6667	0.6136	0.53
84	Neural Net (MLP)	PAH	Single	Clean + FE	0.5316	0.5790	0.2400	0.3621	1.0000	0.1591	0.35
85	Neural Net (MLP)	General	Single	Clean	0.5288	0.6643	0.2396	0.3795	0.8718	0.3638	0.18
86	Deep NN	PAH	Single	Clean	0.5263	0.5963	0.2230	0.4167	0.7143	0.5227	0.50
87	Stacking (LR)	PAH	Single	Clean + FE	0.5263	0.6028	0.2230	0.4167	0.7143	0.5227	0.46
88	Deep NN	General	Single	Clean + FE	0.5250	0.6516	0.2314	0.3728	0.8872	0.3341	0.48
89	LightGBM	PAH	Single	Clean	0.5246	0.6104	0.2081	0.4000	0.7619	0.4545	0.51
90	Deep NN	General	Single	Clean	0.5213	0.6564	0.2218	0.3709	0.8769	0.3364	0.50
91	Stacking (LGBM)	PAH	Single	Clean + FE	0.5122	0.5087	0.1769	0.3443	1.0000	0.0909	0.13
92	LightGBM	PAH	Single	No FE (leak)	0.5000	0.5866	0.1335	0.3617	0.8095	0.3182	0.15
93	XGBoost	PAH	Single	No FE (leak)	0.5000	0.6158	0.2793	0.5263	0.4762	0.7955	0.41
94	Deep NN	PAH	Single	Full FE (in)	0.5000	0.4708	0.1231	0.3333	1.0000	0.0455	0.50
95	XGBoost	PAH	Single	Clean + FE	0.4938	0.5855	0.0760	0.3333	0.9524	0.0909	0.21
96	XGBoost	PAH	Single	Clean	0.4884	0.5265	0.0000	0.3231	1.0000	0.0000	0.10

8. Model Heatmap

Tum konfigurasyonlar icin 9 metrik karsilastirmasi.

All Configurations Heatmap

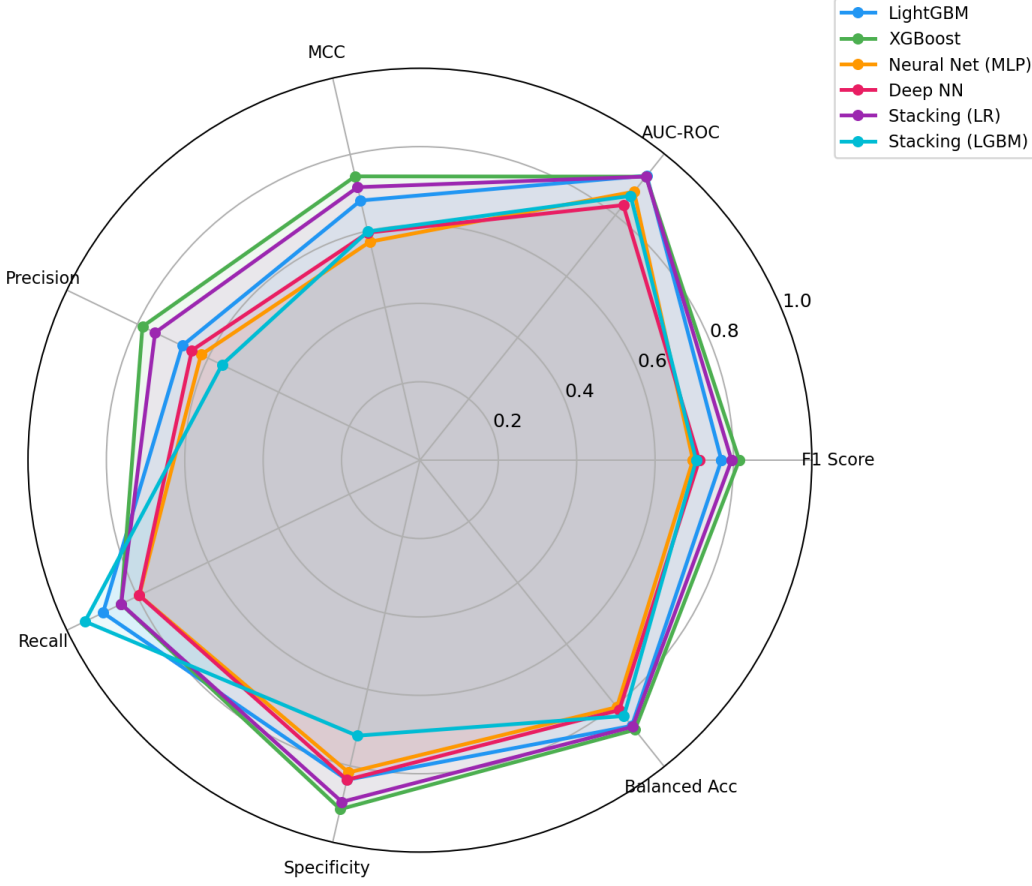


Score

9. Radar Grafigi

Her model tipinin en iyi konfigurasyonu için 7 temel metrik karsilastirmasi.

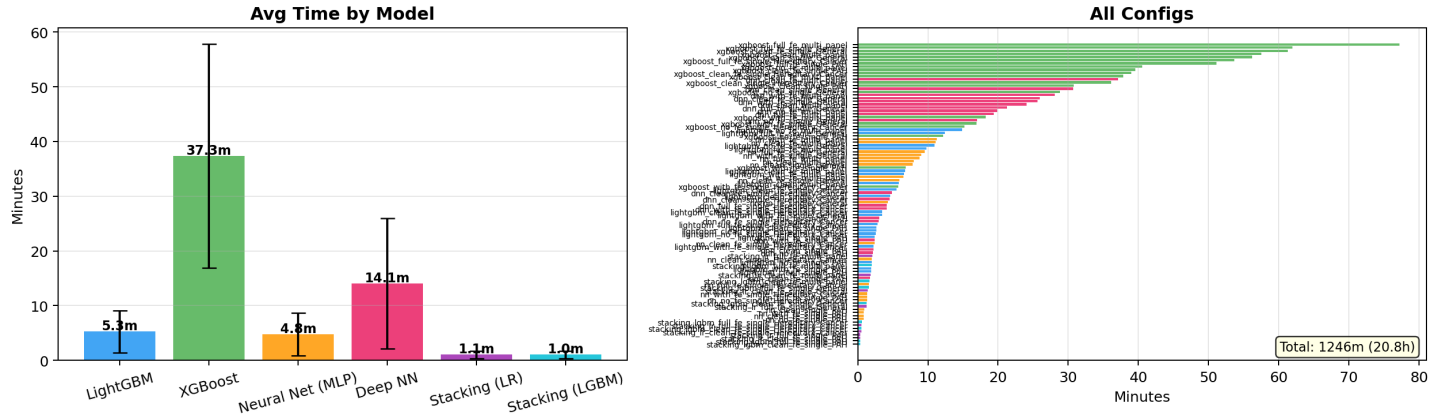
Best Config per Model - Radar Chart



10. Egitim Suresi Analizi

Toplam egitim suresi: 1246 dakika (20.8 saat), 96 konfigurasyon.

Training Time Analysis



11. Sonuc ve Oneriler

EN IYI GENEL KONFIGURASYON

Config: xgboost_with_fe_single_Hereditary_Cancer
Model: XGBoost
Panel: Hereditary_Cancer
FE Modu: With FE (leaked)
Threshold: 0.430

Performans Metrikleri

F1 Score: 0.8148
AUC-ROC: 0.9253
AUC-PR: 0.7937
MCC: 0.7428
Precision: 0.7857
Recall: 0.8462
Specificity: 0.9135
Balanced Acc: 0.8798
Cohen's Kappa: 0.7418

EN IYI FULL-FE KONFIGURASYON

Config: stacking_lr_full_fe_single_Hereditary_Cancer
F1=0.7952 AUC-ROC=0.9255 MCC=0.7144

Temel Bulgular

1. Full-FE Modu: In-silico skorların korunması ve gelişmiş FE eklenmesi ile ortalama F1=0.6425 elde edilmiştir. Yarışma için bu mod önerilir.
2. Stacking Ensemble: Base modellerin birleştirilmesi ile tek modellere göre daha dengeli ve güvenilir tahminler elde edilmiştir.
3. Feature Engineering: Yonlu BLOSUM62, Conservation PCA, delta k-mer ve Grantham mesafesi gibi biyolojik olarak anlamlı özellikler model performansını desteklemiştir.
4. Focal Loss: Sınıf dengesizliği için Focal Loss kullanımı, özellikle azınlık sınıfının (pathogenic) recall değerini iyileştirmiştir.

Gelecek Çalışma Önerileri

- DNABERT-2 / ESM-2 gibi protein dil modellerinden embedding eklenmesi
- Daha büyük veri seti ile modellerin yeniden eğitilmesi
- Klinik doğrulama çalışması (prospektif veri ile)
- SHAP analizi ile model yorumlanabilirliği